



INSURANCE ANALYTICS

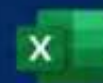
Insurance Analytics Dashboard Project

AGENDA

01		Project Overview	07		Flow Chart for Project
02		Insurance Analytics & It's Importance	08		Excel Dashboards • Policy Dashboard • Branch Dashboard
03		Tools Used for the Project	09		Power BI Dashboards • Policy Dashboard • Branch Dashboard
04		Key Performance Indicators (KPIs)	10		SQL Analysis • Dashboard Outputs • Policy Queries • Branch Queries
05		Dashboard Classification • Policy Dashboard • Branch Dashboard	11		Tableau Dashboards • Policy Dashboard • Branch Dashboard
06		Data Schema	12		Project Outcome & Recommendations
			13		Conclusion



This presentation covers the end-to-end Insurance Analytics project, including data analysis, KPI tracking, dashboard development, and actionable business insights.



Excel



Power BI



SQL



Tableau

Insurance Analytics Project Overview

The Insurance Analytics Project aims to analyse insurance data and transform it into meaningful insights using Excel, SQL, Tableau, and Power BI. The project focuses on understanding policy performance, claim trends, customer behaviour, and branch-wise business performance.

By analysing key metrics such as total policies, premium amount, claim settlements, customer demographics, and branch revenue, the project helps insurance companies make data-driven decisions, improve operational efficiency, and enhance customer satisfaction.

The project involves:

- Data collection and cleaning
- Data analysis using SQL
- Dashboard creation in Excel, Tableau, and Power BI
- Identification of business trends and key performance indicators (KPIs)
- Providing actionable recommendations based on insights

Project Goal:

To convert raw insurance data into interactive dashboards and reports that support better decision-making and business growth.

INSURANCE ANALYTICS & IT'S IMPORTANCE

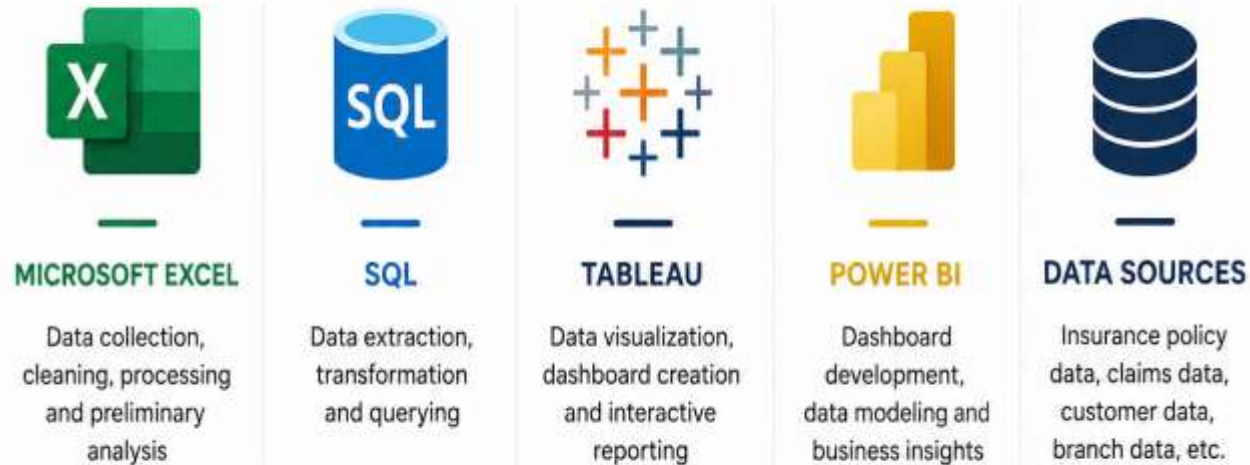
What is Insurance Analytics?

- Collects and analyses insurance-related data to generate meaningful business insights.
- Evaluates policy performance and overall business growth.
- Monitors claims and claim settlement trends.
- Helps understand customer behaviour and demographics.
- Supports risk assessment and business planning.
- Improves operational efficiency through data-driven analysis.
- Enables informed decision-making using key performance indicators (KPIs).
- Enhances overall business performance and profitability

Importance of Insurance Analytics

- Transforms large volumes of insurance data into actionable insights.
- Supports data-driven decision-making and business strategy.
- Helps monitor policy performance and claims management.
- Improves operational efficiency and reporting accuracy.
- Enables identification of customer and market trends.
- Assists in performance tracking through key business KPIs.
- Supports business growth and profitability through informed decisions.

TOOLS USED FOR INSURANCE ANALYTICS



This project utilized a combination of data analysis and visualization tools to transform raw insurance data into meaningful business insights. Microsoft Excel was used for data cleaning, preprocessing, and preliminary analysis. SQL was employed to extract, transform, and manage data efficiently through queries. Tableau was used to create interactive dashboards and visualize key insurance metrics, while Power BI was utilized for advanced reporting, KPI tracking, and business intelligence analysis. Together, these tools enabled effective data analysis, performance monitoring, and data-driven decision-making throughout the project.

KEY POINT INDICATORS FOR INSURANCE ANALYTICS DASHBOARD PROJECT(KPI'S)

 **Total
Policies**

 **Total
Customers**

 **Age
Bucket-wise
Policy Count**

 **Gender-
wise Policy
Count**

 **Policy
Type-wise
Count**

 **Policy
Expiry This
Year**

 **Growth
%**

 **Claim
Status-wise
Policy Count**

 **Payment
Status-wise
Policy Count**

 **Total
Claim
Amount**

DASHBOARD CLASSIFICATION

```
graph TD; A[DASHBOARD CLASSIFICATION] --> B[POLICY DASHBOARD]; A --> C[BRANCH DASHBOARD]; A --> D[Policy Dashboard<br/>• Policy Performance Analysis<br/>• Claims Analysis<br/>• Customer Demographics<br/>• Payment & Renewal Tracking<br/>Branch Dashboard<br/>• Target vs Achievement Analysis<br/>• Sales Performance Monitoring<br/>• Opportunity Management<br/>• Revenue & Business Growth Analysis];
```

POLICY DASHBOARD

BRANCH DASHBOARD

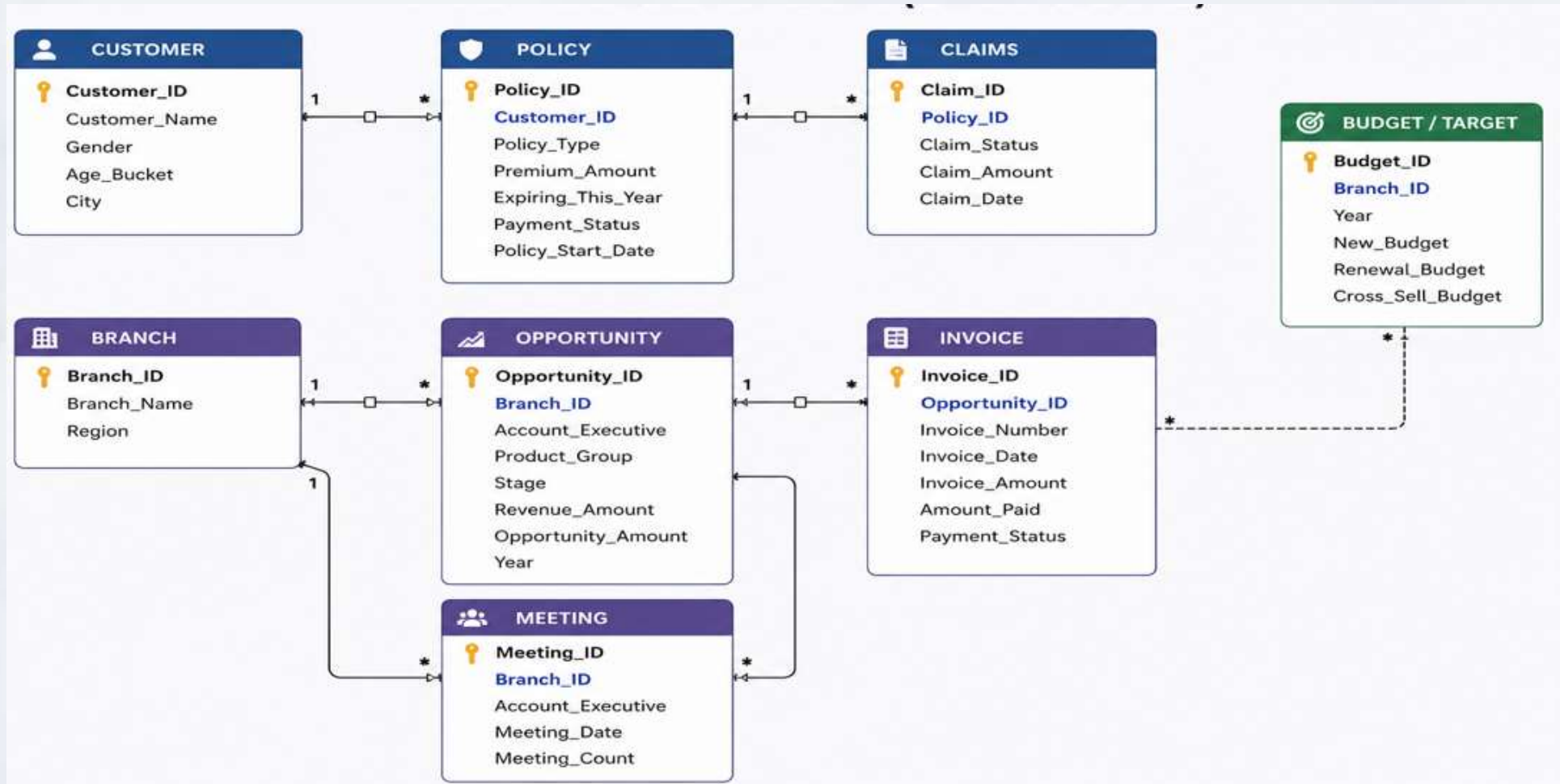
Policy Dashboard

- Policy Performance Analysis
- Claims Analysis
- Customer Demographics
- Payment & Renewal Tracking

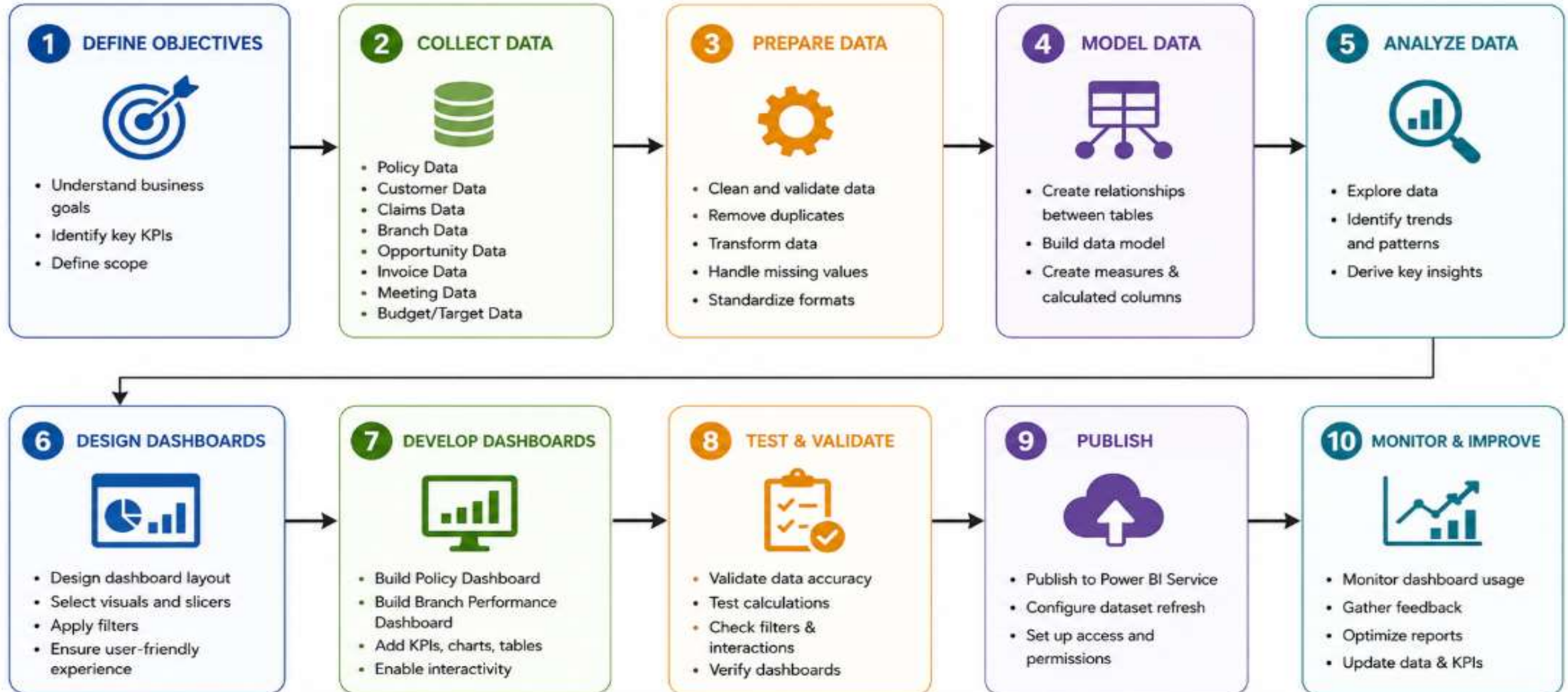
Branch Dashboard

- Target vs Achievement Analysis
- Sales Performance Monitoring
- Opportunity Management
- Revenue & Business Growth Analysis

DATA SCHEMA MODEL



INSURANCE ANALYTICS – FLOW CHART



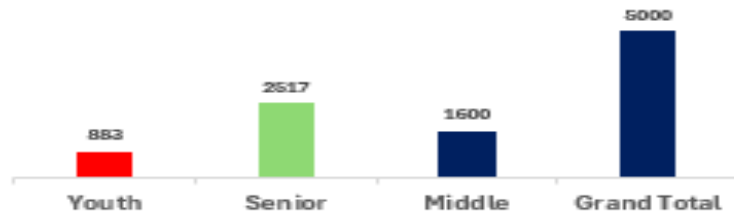
EXCEL POLICY DASHBOARD

Insurance Policy Dashboard

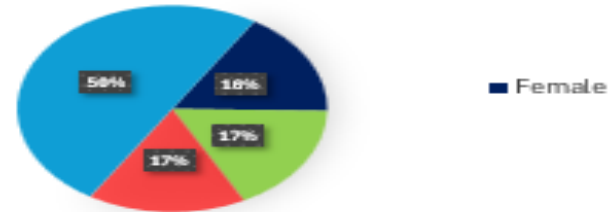
Total Policy 5000

Total Customer 3148

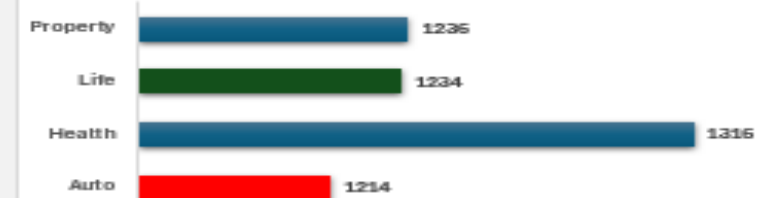
Age Bucket wise policy count



Gender wise policy count



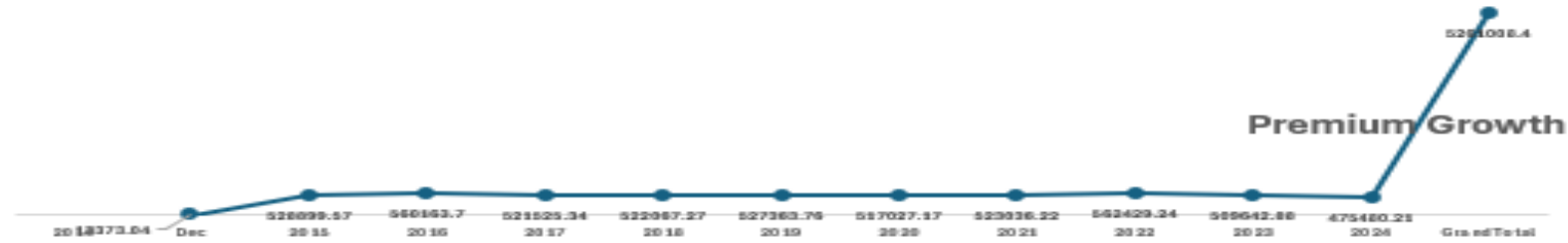
Policy type wise count



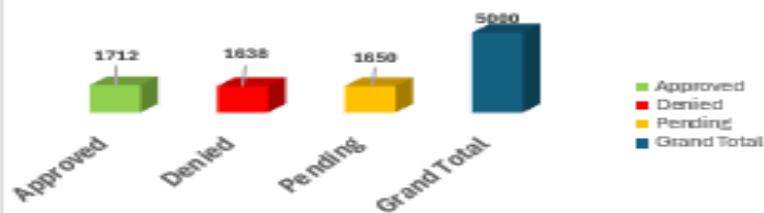
Policy Expire



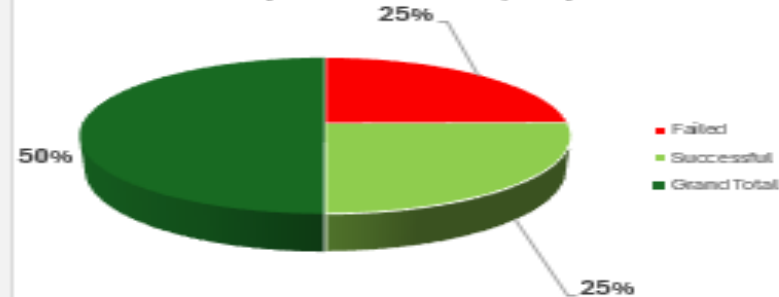
Premium Growth



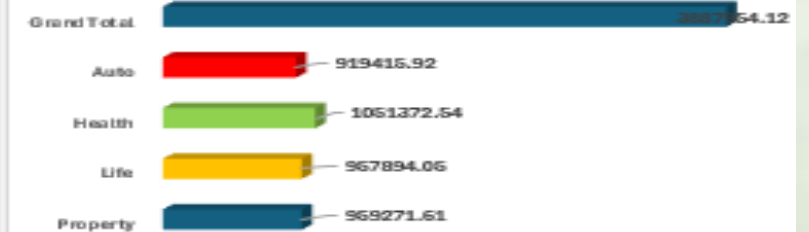
Claim status policy



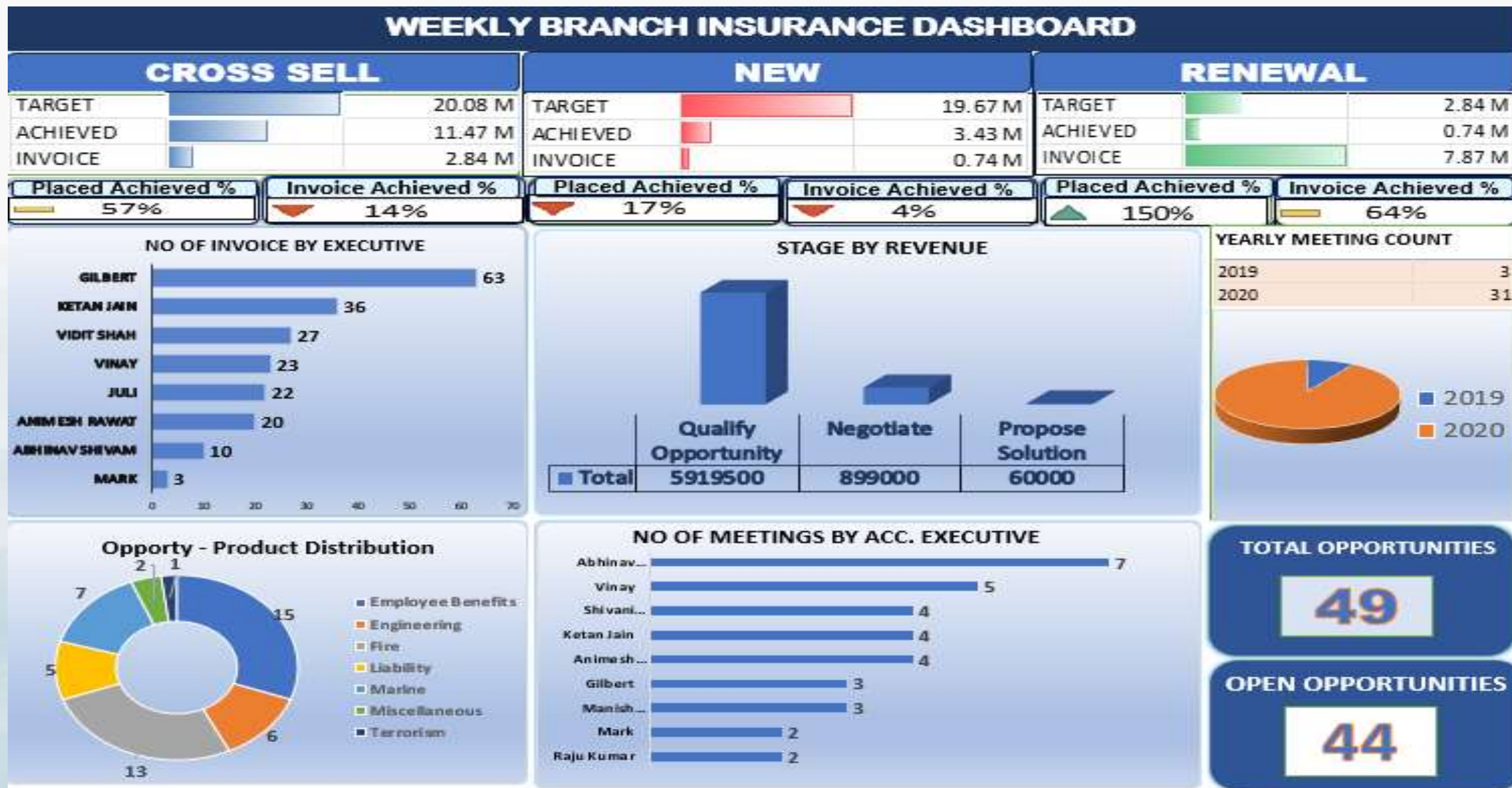
Payment status wise policy



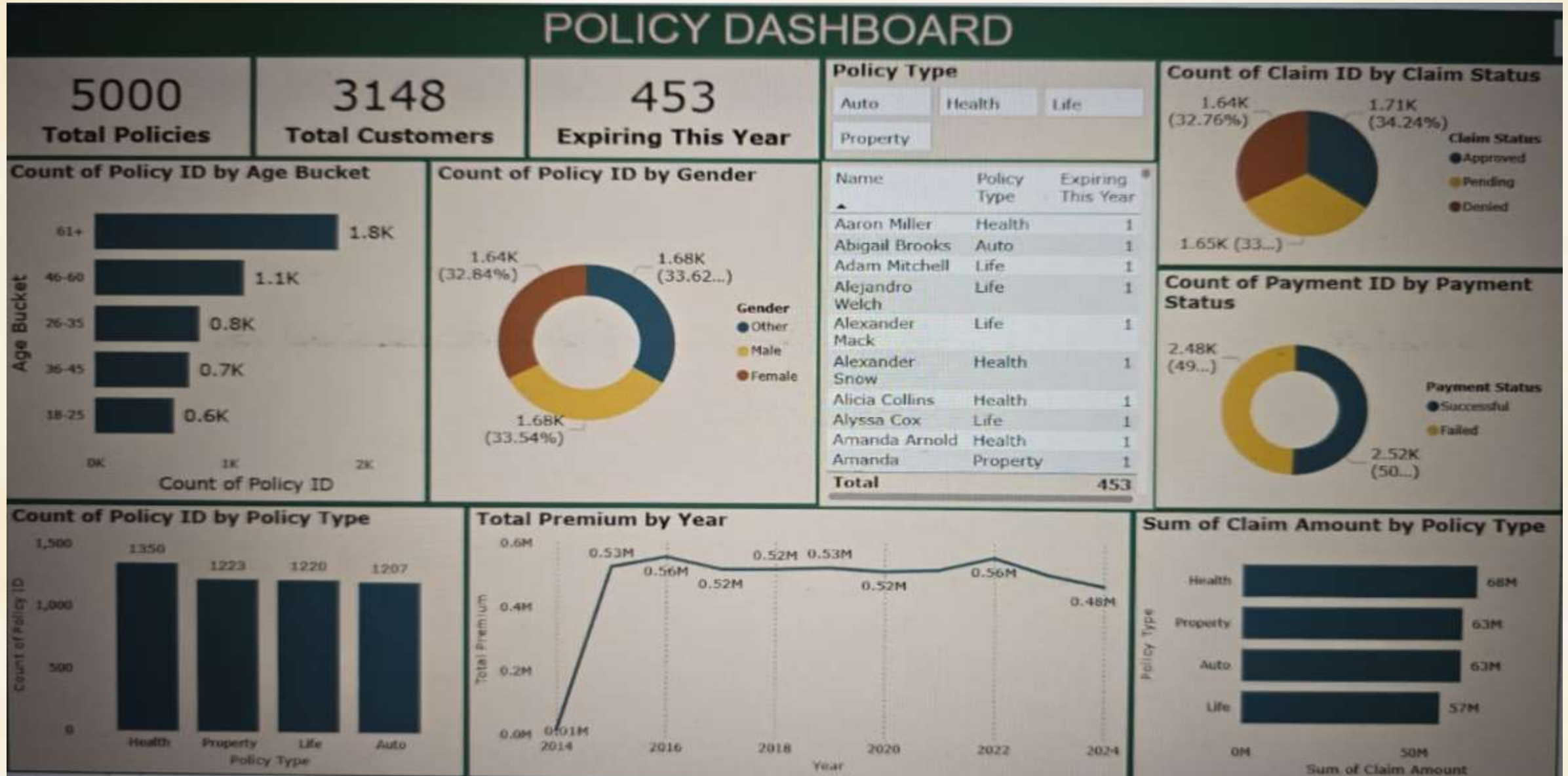
Total claim amount policy type



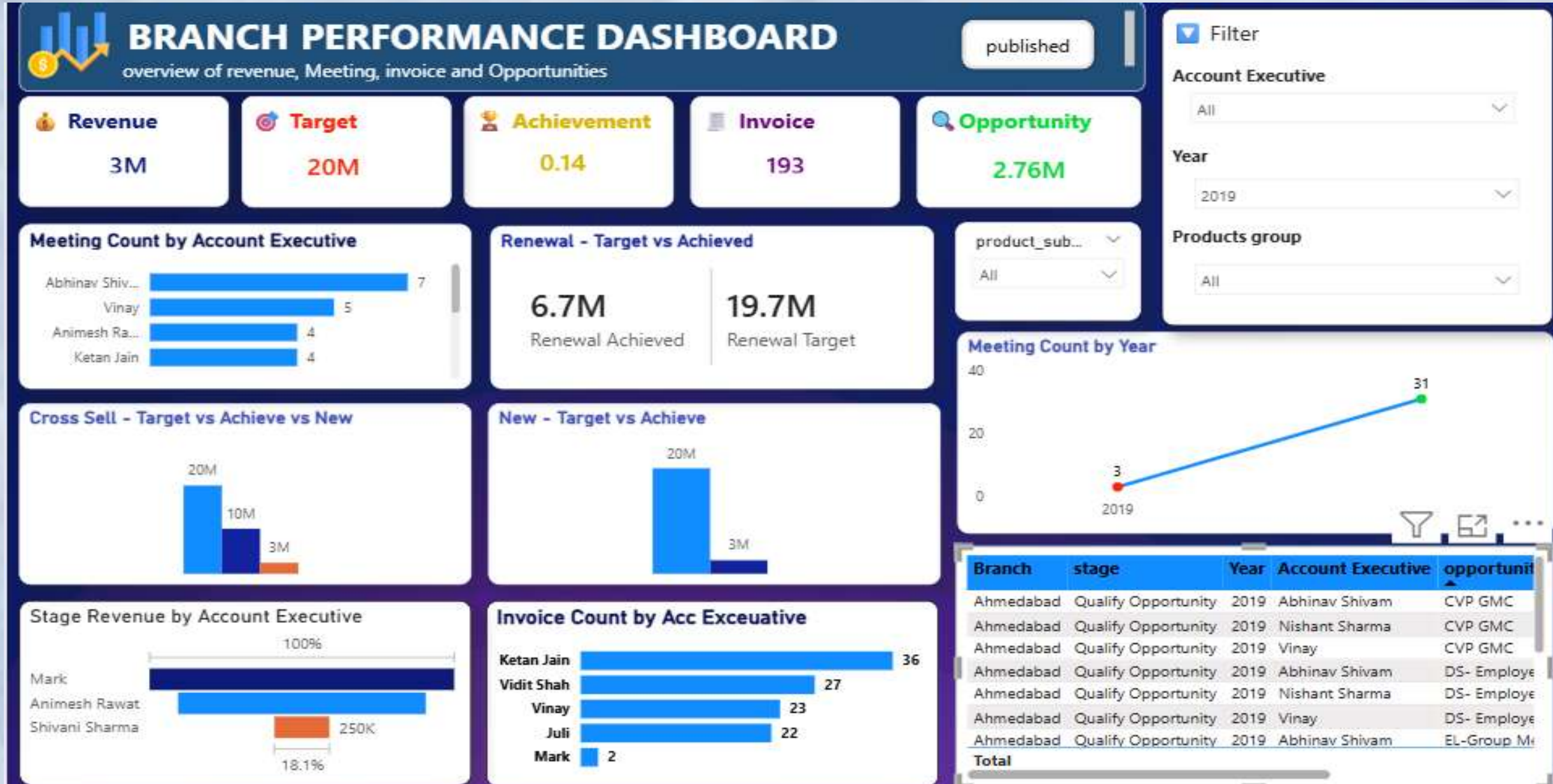
EXCEL BRANCH DASHBOARD



POWER BI POLICY DASHBOARD



POWER BI BRANCH DASHBOARD



SQL POLICY OUTPUTS

1. TOTAL POLICY

Total_Policy

56915

2. TOTAL CUSTOMERS

Total_Customers

35162

6. POLICIES EXPIRING THIS YEAR

Policies_Expiring_This_Year

8243

10. TOTAL CLAIM AMOUNT

Total_Claim_Amount

18507272.64

3. AGE BUCKET WISE POLICY COUNT

Age Group	Policy_Count
-----	-----
18-25	3615
26-35	12842
36-45	15327
46-55	13429
56-65	8276
65+	3426

4. GENDER WISE POLICY COUNT

Gender	Policy_Count
-----	-----
Male	33560
Female	22877
Other	478

5. POLICY TYPE WISE POLICY COUNT

Policy Type	Policy_Count
-----	-----
Health	22983
Auto	14276
Life	11302
Property	8354

7. PREMIUM GROWTH RATE

Year	Total_Premium
-----	-----
2019	8244310.00
2020	11807200.00
2021	143056450.00
2022	167542200.00
2023	201536900.00
2024	221021110.00

8. CLAIM STATUS WISE POLICY COUNT

Claim Status	Policy_Count
-----	-----
Approved	30130
Pending	17430
Denied	9355

9. PAYMENT STATUS WISE POLICY COUNT

Payment Status	Policy_Count
-----	-----
Successful	49892
Failed	7023

SQL BRANCH OUTPUTS

1. CROSS SELL TARGET		2. CROSS SELL ACHIEVED		3. CROSS SELL PLACED ACHIEVEMENT %		4. CROSS SELL INVOICE		5. CROSS SELL INVOICE ACHIEVEMENT %			
CrossSell Target		CrossSell Achieved		Crss Sell Plcd Achvmnt %		Cross Sell Invoice		Crss Sell Invoice Achvmnt %			
85.00 M		62.75 M		73.82% ▼		58.40 M		68.71% ▼			
6. NEW TARGET		7. NEW ACHIEVED		8. NEW PLACED ACHIEVEMENT %		9. NEW INVOICE		10. NEW INVOICE ACHIEVEMENT %			
New Target		New Achieved		New Plcd Achvmnt %		New Invoice		New Invoice Achvmnt %			
120.00 M		98.30 M		81.92% ▼		90.15 M		75.13% ▼			
11. RENEWAL TARGET		12. RENEWAL ACHIEVED		13. RENEWAL PLACED ACHIEVEMENT %		14. RENEWAL INVOICE		15. RENEWAL INVOICE ACHIEVEMENT %			
Renewal Target		Renewal Achieved		Renewal Plcd Achvmnt %		Renewal Invoice		Renewal Invoice Achvmnt %			
200.00 M		185.07 M		92.54% ▼		178.60 M		89.30% ▼			
16. MEETING COUNT BY YEAR			17. OPPORTUNITIES SUMMARY			18. OPPORTUNITY BY REVENUE – TOP 4					
2019 Meeting Count		2020 Meeting Count		Total Opportunities		Total Open Opportunities		Opportunity Name		Oppty by Revenue-Top 4	
								EL-Group Mediclaim		40.00 M	
312		289		49		44		DB -Mega Policy		40.00 M	
								CVP GMC		35.00 M	
								DS- Employees GMC		30.00 M	
19. NO OF MEETING BY ACCOUNT EXECUTIVE		20. NO OF INVOICE BY ACCOUNT EXECUTIVE				21. STAGE BY REVENUE			22. OPPORTUNITY PRODUCT DISTRIBUTION		
Account Executive	No_of_Meeting_by_Accnt_Exec	Account Executive	Income Class	No of Invoice by Accnt Exec	Total Invoice Count	Stage	Revenue	Stage by Revenue	Product Group	Oppty-Product distribution	
Abhinav Shivam	7	Abhinav Shivam	Renewal	15	28	Qualify Opportunity	59.19 M	32.68 %	Employee Benefits	15	
			Cross Sell	10	28	Negotiate	89.90 M	49.62 %	Fire	13	
			New	3	28	Propose Solution	60.00 M	33.10 %	Marine	7	
Vinay	5	Vinay	Renewal	12	23	Closed Won	120.50 M	66.54 %	Engineering	6	
			Cross Sell	7	23	Closed Lost	15.30 M	8.44 %	Liability	5	
			New	4	23				Miscellaneous	2	
Ankita Shah	4	Ankita Shah	Renewal	6	16				Terrorism	1	
Aditya Shah	4		Cross Sell	5	16						
Wati Shah	3		New	3	16						
Gautam Murlunde	2					23. OPEN OPPORTUNITIES – TOP 7					
Shruti Sharma	2					Opportunity Name		Open Oppty-Top 7			
						EL-Group Mediclaim		40.00 M			
						DB -Mega Policy		40.00 M			
						CVP GMC		35.00 M			
						DS- Employees GMC		30.00 M			
						FM-Group Mediclaim		30.00 M			
						Icy Shield		25.00 M			
						Secure Health		22.50 M			

SQL POLICY QUERY

policy script x

Limit to 1000 rows

```
1 • USE INSURANCE_PROJECT;
2 #1.TOTAL POLICY
3 • SELECT COUNT(*) AS Total_Policy FROM policy_details;
4
5 #2. TOTAL CUSTOMERS
6 • SELECT COUNT(DISTINCT `Customer ID`) AS Total_Customers FROM policy_details;
7
8 #3.AGE BUCKET WISE POLICY COUNT
9
10
11 • SELECT c.`Age Group`, COUNT(*) AS Policy_Count FROM customer_info c
12 JOIN policy_details p
13 ON c.`Customer ID` = p.`Customer ID` GROUP BY c.`Age Group`;
14
15 #4. GENDER WISE POLICY COUNT
16 • SELECT c.Gender, COUNT(*) AS Policy_Count FROM customer_info c
17 JOIN policy_details p
18 ON c.`Customer ID` = p.`Customer ID` GROUP BY c.Gender ORDER BY Policy_Count DESC;
19
20 #5.POLICY TYPE WISE POLICY COUNT
21 • SELECT `Policy Type`, COUNT(*) AS Policy_Count
22 FROM policy_details GROUP BY `Policy Type` ORDER BY Policy_Count DESC;
23
24 #6.POLICY EXPIRE THIS YEAR
25 • SELECT COUNT(*) AS Policies_Expiring_This_Year
26 FROM policy_details
27 WHERE YEAR(STR_TO_DATE(`Policy End Date`, '%d-%m-%Y')) = YEAR(CURDATE());
28
29 #7.PREMIUM GROWTH RATE
```

```
30 • SELECT YEAR(STR_TO_DATE(`Policy Start Date`, '%d-%m-%Y')) AS Year, ROUND(SUM(`Premium Amount`),2) AS Total_Premium
31 FROM policy_details
32 GROUP BY YEAR(STR_TO_DATE(`Policy Start Date`, '%d-%m-%Y'))
33 ORDER BY Year;
34
35 #8.CLAIM STATUS WISE POLICY COUNT
36 • SELECT `Claim Status`, COUNT(*) AS Policy_Count FROM claims GROUP BY `Claim Status` ORDER BY Policy_Count DESC;
37
38 #9.PAYMENT STATUS WISE POLICY COUNT
39 • SELECT `Payment Status`, COUNT(`Policy ID`) AS Policy_Count FROM payment_history GROUP BY `Payment Status`
40 ORDER BY Policy_Count DESC;
41
42 #10. TOTAL CLAIM AMOUNT
43 • SELECT ROUND(SUM(`Claim Amount`),2) AS Total_Claim_Amount FROM claims;
```

SQL BRANCH QUERY

new insurance script x

Limit to 1000 rows

```
1
2 ● USE INSURANCE_PROJECT;
3
4 /*CROSS SELL*/
5 ● SELECT concat(format(sum('Cross Sell Budget')/1000000,2),' M') AS 'CrossSell Target' FROM budget;
6 ● SELECT concat(format((((SELECT SUM(Amount) FROM brokerage WHERE income_class = 'Cross sell')+(SELECT SUM(Amount) FROM fees WHERE income_class = 'Cross sell'))/1000000,2),' M') AS 'CrossSell Achieved';
7 ● SELECT concat(format((((SELECT SUM(Amount) FROM brokerage WHERE income_class='Cross sell')+(SELECT SUM(Amount) FROM fees WHERE income_class = 'Cross sell'))/(SELECT SUM('Cross Sell Budget') FROM budget)*100,2),'% ',case when ((SELECT SUM(Amount) FROM brokerage WHERE income_class='Cross
8 ● SELECT concat(format(SUM(Amount)/1000000,2),' M') AS 'Cross Sell Invoice' FROM Invoice WHERE income_Class='Cross Sell';
9 ● SELECT concat(format((SELECT SUM(Amount) FROM Invoice WHERE income_Class='Cross Sell')/(SELECT SUM('Cross Sell Budget') FROM budget)*100,2),'% ',case when ((SELECT SUM(Amount) FROM Invoice WHERE income_Class='Cross Sell')/(SELECT SUM('Cross Sell Budget') FROM budget))<1 then '▼' else '▲'
10
11 /*NEW*/
12 ● SELECT concat(format(SUM('New Budget')/1000000,2),' M') AS 'New Target' FROM budget;
13 ● SELECT concat(format((((SELECT SUM(Amount) FROM brokerage WHERE income_class = 'New')+(SELECT SUM(Amount) FROM fees WHERE income_class = 'New'))/1000000,2),' M') AS 'New Achieved';
14 ● SELECT concat(format((((SELECT SUM(Amount) FROM brokerage WHERE income_class='New')+(SELECT SUM(Amount) FROM fees WHERE income_class = 'New'))/(SELECT SUM('New Budget') FROM BUDGET)*100,2),'% ',case when (((SELECT SUM(Amount) FROM brokerage WHERE income_class='New')+(SELECT SUM(Amount) F
15 ● SELECT concat(format(SUM(Amount)/1000000,2),' M') AS 'New Invoice' FROM Invoice WHERE income_Class='New';
16 ● SELECT concat(format((SELECT SUM(Amount) FROM Invoice WHERE income_Class='New')/(SELECT SUM('New Budget') FROM budget)*100,2),'% ',case when ((SELECT SUM(Amount) FROM Invoice WHERE income_Class='New')/(SELECT SUM('New Budget') FROM budget))<1 then '▼' else '▲' end) AS 'New Invoice Achvm
17
18 /*RENEWAL*/
19 ● SELECT concat(format(SUM('Renewal Budget')/1000000,2),' M') AS 'Renewal Target' FROM budget;
20 ● SELECT concat(format((((SELECT SUM(Amount) FROM brokerage WHERE income_class = 'Renewal')+(SELECT SUM(Amount) FROM fees WHERE income_class = 'Renewal'))/1000000,2),' M') AS 'Renewal Achieved';
21 ● SELECT concat(format((((SELECT SUM(Amount) FROM brokerage WHERE income_class='Renewal')+(SELECT SUM(Amount) FROM fees WHERE income_class = 'Renewal'))/(SELECT SUM('Renewal Budget') FROM budget)*100,2),'% ',case when (((SELECT SUM(Amount) FROM brokerage WHERE income_class='Renewal')+(SELE
22 ● SELECT concat(format(SUM(Amount)/1000000,2),' M') AS 'Renewal Invoice' FROM Invoice WHERE income_Class='Renewal';
23 ● SELECT concat(format((SELECT SUM(Amount) FROM Invoice WHERE income_Class='Renewal')/(SELECT SUM('Renewal Budget') FROM budget)*100,2),'% ',case when ((SELECT SUM(Amount) FROM Invoice WHERE income_Class='Renewal')/(SELECT SUM('Renewal Budget') FROM budget))<1 then '▼' else '▲' end) AS 'R
24
25 /*YEARLY MEETING COUNT*/
26 ● SELECT (SELECT count(meeting_date) from meeting where year(DATE_FORMAT(STR_TO_DATE(meeting_date, '%d-%m-%Y'), '%Y-%m-%d'))=2019) as '2019 meeting count',(SELECT count(meeting_date) from meeting where year(DATE_FORMAT(STR_TO_DATE(meeting_date, '%d-%m-%Y'), '%Y-%m-%d'))=2020) as '2020 mee
27
28 /*No of Meeting by Acctn Exec*/
29 ● SELECT 'Account Executive',count(meeting_date) AS No_of_Meeting_by_Acctn_Exec from meeting group by 'Account Executive' order by 2 desc;
30
31
32 /*No of Invoices by Acctn Exec*/
33
34 #ALTER TABLE INVOICE CHANGE COLUMN 'InVOICE_NUMBER' 'INVOICE_NUMBER' TEXT;
35
36 ✖ WITH InvoiceCounts AS (SELECT 'Account Executive',income_class,COUNT(invoice_number) AS invoice_count FROM Invoice GROUP BY 1,2),AccountExecCounts AS (SELECT 'Account Executive',SUM(invoice_count) AS total_invoice_count FROM InvoiceCounts GROUP BY 1) SELECT ic.'Account Executive',ic.ino
37
38 /*Opportunities*/
39
```

39

40

41 ● SELECT (SELECT count(opportunity_name) from Opportunity) AS 'Total Opportunities',(SELECT count(opportunity_name) from Opportunity where (stage='Qualify Opportunity' or stage='Propose Solution')) AS 'Total Open Opportunities';

42

43 /*Oppty by Revenue-Top 4*/

44 ● SELECT opportunity_name,concat(format(sum(revenue_amount)/1000000,2),' M') AS 'Oppty by Revenue-Top 4' from Opportunity group by 1 order by 2 desc limit 4;

45

46 /*Stage by Revenue*/

47 ● SELECT stage,concat(format(sum(revenue_amount)/1000000,2),' M') as Revenue,concat(format((sum(revenue_amount)/(Select sum(revenue_amount) from Opportunity))*100,2),' %') AS 'Stage by Revenue' from Opportunity group by 1;

48

49 /*Oppty-Product distribution*/

50 ● SELECT product_group,count(opportunity_name) AS 'Oppty-Product distribution' from Opportunity group by 1 order by 2 desc;

51

52 /*Open Oppty-Top 4*/

53 ● SELECT opportunity_name,concat(format(sum(revenue_amount)/1000000,2),' M') AS 'Open Oppty-Top 4' from Opportunity where (stage='Qualify Opportunity' or stage='Propose Solution') group by 1 order by 2 desc limit 7;

TABLEAU POLICY DASHBOARD

Tableau Insurance Policy Dashboard

Total Claim Amount				
Policy Type				
Auto	Health	Life	Property	
568,089	697,673	599,349	580,709	

Total Policy

5,000

Total Customers

3,148

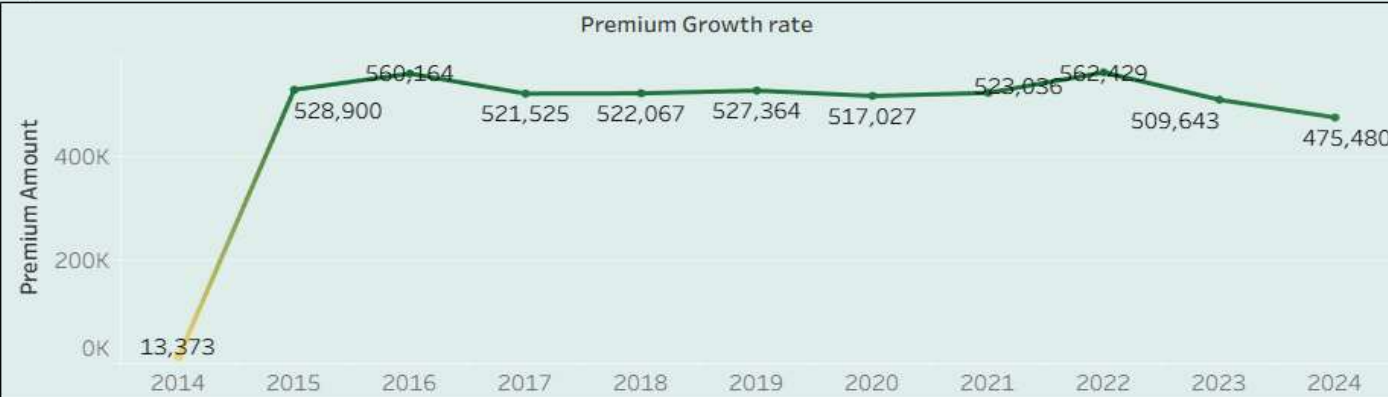
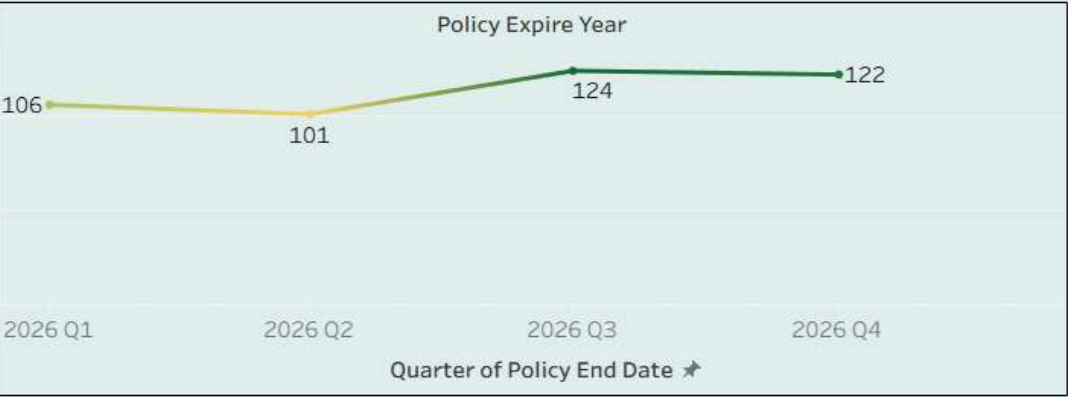
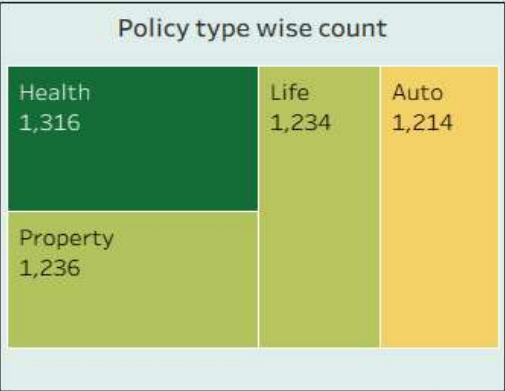
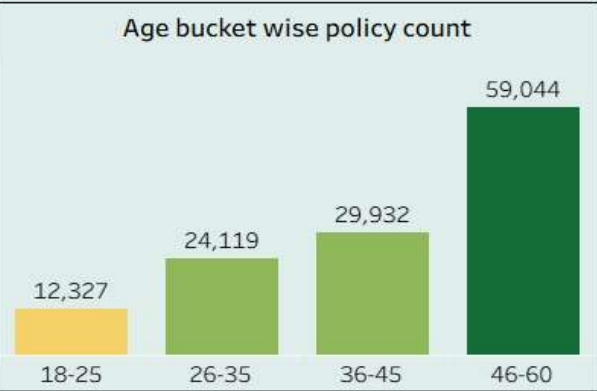
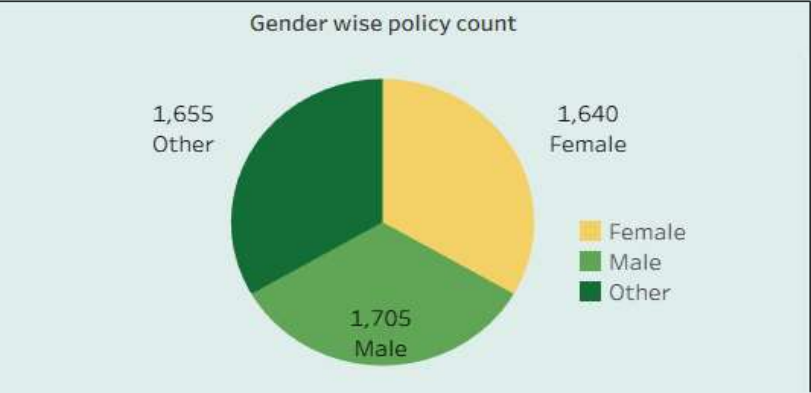


TABLEAU BRANCH DASHBOARD



Insurance Branch Performance Dashboard



Invoice
193



Meeting
34



FY Target
52.1M



Achievement
33.4M



Achievement
64.1%



Opportunity
49



Cross Sell - Target vs Achieved



New Business - Target vs Achieved



Renewal Achieved
18.5M



Renewal Target
12.3M



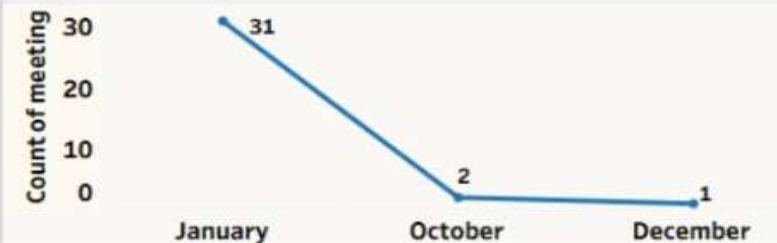
Stage Funnel by Revenue



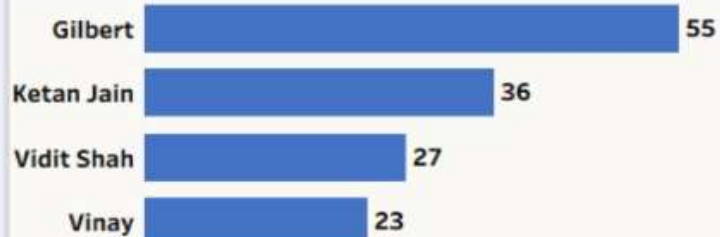
Top Open Opportunities



Monthly Meeting Trend



Invoice Count by Account Executive



Meeting Count by Account Executive



Opportunity Stage Distribution



PROJECT OUTCOME:

This Insurance Analytics Dashboard project converted raw insurance data into meaningful insights for better decision-making. It provided visibility into key metrics such as total policies, premium collected, revenue contribution, and cross-sell performance, helping management track overall business performance and targets.

It also enabled branch-wise analysis to identify both high-performing and underperforming branches, allowing better focus on improvement areas and efficiency optimization. In addition, cross-sell analysis highlighted opportunities to increase revenue from existing customers, improving profitability without acquiring new customers.

Overall, the dashboard reduced manual reporting, saved time, and improved data transparency by bringing all important insights into a single view. It supports faster, data-driven decisions and better business planning for growth.

RECOMMENDATIONS:

Based on the insights from the dashboard, the following recommendations can be made to improve business performance and growth:

The company should focus on improving **cross-sell strategies** by targeting existing customers with personalized insurance products, as this can increase revenue without additional customer acquisition cost. Underperforming branches should be given special attention through training, better resources, and performance monitoring to improve their efficiency and output.

Regular monitoring of key metrics like policies, premium, and revenue will help track performance effectively. Maintaining an updated dashboard will support faster and data-driven decision-making, ultimately improving overall business growth and profitability.

CONCLUSION:

This Insurance Analytics Dashboard project was successfully completed to analyse and visualize key insurance business data.

This Insurance Analytics Dashboard project was developed to simplify and analyse insurance business data in an effective way. It helped in understanding important metrics such as policies, premium, revenue, and cross-sell performance in a simple and effective way.

The dashboard provided better visibility into branch performance and supported easy comparison between different branches. It also helped identify areas for improvement and opportunities for increasing revenue.

Overall, the project made data easier to understand, reduced manual analysis, and supported faster decision-making for business growth.

